Composition-Aware Cycle-Consistent Domain Translation for Single-Cell and Bulk Transcriptomics

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1. Overview

Real biological datasets are rarely collected the same way every time. Different measurement batches, instruments, or protocols introduce systematic variation mixed in with the biological signal of interest. In genomics, this technical disparity is called a batch effect; resolving it means removing the non-biological variance without losing the real signal underneath.

We focus on one clear, structured batch effect, studied here in a paired dataset of single-cell and bulk RNA-seq from the same blood draws. RNA sequencing quantifies gene activity within a sample: bulk RNA-seq averages expression across every cell in the sample, whereas single-cell RNA-seq (scRNA-seq) profiles individual cells at higher resolution. In this cohort, single-cell RNA-seq is performed on PBMC (peripheral blood mononuclear cells) while bulk RNA-seq uses whole blood — different preparations from the same blood draw, so despite identical donor biology, the profiles still look different in a consistent way. This difference comes mainly from cell-type composition: PBMC isolation physically depletes granulocytes (50–70% of whole blood), which bulk assays retain. CytoSet [8], a permutation-invariant set encoder that pools single cells for sample-level prediction, is trained on single-cell data in this cohort to classify donors as young or old. Its embedding space is organized by assay identity instead of by age, so predictions do not transfer when the same encoder is applied to bulk profiles. Standard batch-correction methods (ComBat, CORAL) pull the single-cell and bulk embeddings closer together in that space (Figure 1). This correction distorts the non-linear biological signal itself instead of removing only the technical difference, so transfer does not recover. We frame this project as domain-to-domain translation: learning a non-linear, cycle-consistent adversarial mapping (CyCADA-style). The mapping is supervised by a task-consistency loss tied directly to real, measured immune-cell composition.

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2. Related Work

CycleGAN [9] introduced cycle-consistency for unpaired image translation. CyCADA [2] added task-consistency for domain adaptation, but targeted vision classification with discrete cross-entropy objectives. We therefore replace CyCADA's classification loss with a continuous regression loss, applied to 5000-dimensional gene expression vectors instead of images. Standard genomics baselines ComBat [3] and CORAL [5] apply linear corrections that cannot model asymmetric cell-type depletion; neither closes the gap in our evaluation (Figure 1). Recent transcriptomic GANs apply to different settings: GANomics [7] bridges microarray and RNA-seq platforms on identical tissue extracts rather than distinct cell preparations from the same draw. STCGAN [6] pretrains a cycle-consistent GAN, then estimates composition afterward in a separate module, two decoupled stages. Our approach differs: composition is the task-consistency signal inside the generator's own training loop from the start, one objective, not two.

3. Approach

Both domains are represented as 5000-dimensional vectors, log-normalized to a common total count per sample (log-CP10k, a standard RNA-seq scaling), over a shared gene panel (the genes reliably detected by both assays), so each dimension refers to the same gene in both domains. Single-cell data is aggregated into sample-level pseudobulk, $x_S \in \mathcal{S}$, by summing counts per sample so the result resembles a real bulk profile. Bulk data is represented as $x_B \in \mathcal{B}$. Operating on pseudobulk rather than individual cells preserves a fixed-dimensional metric space, needed for discriminator stability and tractable ℓ_1 cycle-consistency. Modeling raw single cells would instead require set-level discriminators and set-to-set distances (e.g., optimal transport), which are notoriously difficult to train reliably; bulk profiles also lack cell-level resolution to target directly. Pseudobulk is also the representation on which cross-domain signal survives, a main result from our ongoing CytoSet benchmarking study. The set encoder configuration (Overview) induces collapse instead.

Generators $G_{S\to B}$ and $G_{B\to S}$ are parameterized as

residual MLPs ($G(x) = x + \text{MLP}(x)$, $5000 \to 512 \to 512 \to 5000$). The residual principle is established, echoing how CycleGAN's own generators avoid reconstructing from scratch. Applying it as a flat MLP over tabular gene-expression vectors, rather than CycleGAN's convolutional image architecture, is our own design choice. ComBat and CORAL are both linear corrections, so neither can capture non-linear structure in the shift between domains; the learned MLP adds that non-linear correction on top. Keeping the correction additive, instead of generating the output from scratch, keeps the generator anchored to the real input and unable to destroy real biological signal while correcting it.

Discriminators D_S and D_B are MLPs trained with the established LSGAN objective [4]. The $S \to B$ direction is optimized as $\min_{G_{S\to B}} \max_{D_B}$:

$$\mathcal{L}_{\text{adv}}^{S\to B} = \mathbb{E}_{x_B}[(D_B(x_B)-1)^2] + \mathbb{E}_{x_S}[D_B(G_{S\to B}(x_S))^2], \tag{1}$$

A symmetric $\mathcal{L}_{\text{adv}}^{B\to S}$ covers the reverse direction, and the total adversarial term used below is $\mathcal{L}_{\text{adv}} = \mathcal{L}_{\text{adv}}^{S\to B} + \mathcal{L}_{\text{adv}}^{B\to S}$. Cycle-consistency follows CycleGAN [9] directly:

$$\begin{aligned}\mathcal{L}_{\text{cyc}} = \mathbb{E}_{x_S}[\|G_{B\to S}(G_{S\to B}(x_S)) - x_S\|_1] \\ + \mathbb{E}_{x_B}[\|G_{S\to B}(G_{B\to S}(x_B)) - x_B\|_1].\end{aligned} \tag{2}$$

Both of the above are established machinery, unchanged from prior work.

This is where our approach departs from CyCADA. Instead of a classifier trained on a proxy label, the task loss uses two regressors, C_{bulk} and C_{sc}, which predict cell-type composition (the fraction of a sample's cells in each major immune lineage, such as T cells, B cells, or monocytes) as a vector on the simplex Δ^{K-1}, computed directly from established single-cell annotations. Matched pairs originate from the same draw, so one invariant ground-truth composition vector, c^*, supervises both directions without deconvolution. c^* is measured directly from real biology, not predicted by a model. This is the point where biology drives the objective directly. Both task networks are pretrained to convergence and frozen. Our mathematical contribution is this task-consistency term:

$$\mathcal{L}_{\text{task}} = \|C_{\text{bulk}}(G_{S\to B}(x_S)) - c^*\|_1 + \|C_{\text{sc}}(G_{B\to S}(x_B)) - c^*\|_1. \tag{3}$$

The full objective follows CyCADA's [2] structure, with this term substituted for its classification-based one: $\mathcal{L} = \mathcal{L}_{\text{adv}} + \lambda_{\text{cyc}}\mathcal{L}_{\text{cyc}} + \lambda_{\text{task}}\mathcal{L}_{\text{task}}$.

4. Experiments

Data: We evaluate on AIFI Sound Life [1]: 96 healthy donors, each with a matched single-cell (on PBMC) and bulk (on whole-blood) sample pair from the same draw, for ∼840 pairs total (public, CC BY 4.0). This same-draw matching ensures the domain gap reflects measurement assay differences, not donor identity or draw timing. Splits are donor-grouped, since donors contribute multiple samples; preprocessing and loaders are pre-constructed. At ∼840 pairs, this is a modest sample size for GAN training — a real limitation on what the evaluation below can establish.

Baselines & Evaluation: We benchmark against uncorrected data, ComBat, and CORAL, and report: **(1) Domain Alignment** — reductions in the silhouette score (how separated the two domains are in the embedding, where lower is better) and in how much of the top principal component reflects modality, not biology; **(2) Downstream Transfer** — evaluating corrected profiles on CytoSet's original task, with its weights frozen, classifying donors as young or old, scored by AUROC; **(3) Translation Accuracy** — per-sample correlation (R^2, Spearman ρ) with the real paired profile, plus per-gene mean/variance agreement; **(4) Task Generalization** — composition-prediction accuracy on donors held out from both task-net pretraining and GAN training, testing whether the signal generalizes rather than being memorized; and **(5) Sanity Checks** — per-sample output diversity, held-out cycle-reconstruction error, and whether marker genes (e.g., CD3D/CD3E for T cells) still show the expected pattern after translation.

Ablations: Each ablation tests a specific design decision from Approach rather than sweeping an arbitrary hyperparameter, ordered from most fundamental to most specific: **(1) No Task Loss** — training with only the adversarial and cycle-consistency terms ($\lambda_{\text{task}} = 0$), the plain CycleGAN case, to establish whether task-consistency contributes anything beyond cycle-consistency alone; **(2) Task Loss Type** — swapping C_{bulk}/C_{sc} for CytoSet's frozen young-vs-old classifier, CyCADA's original classification-based formulation, to test whether real biology grounds the task loss better than a generic signal; and **(3) Generator Architecture** — comparing the residual generator ($G(x) = x + \text{MLP}(x)$) against a non-residual MLP of the same capacity, to test whether the additive property is actually what protects biological signal during translation, as Approach claims.

5. Others

We want to know two things: whether our translation approach can close the single-cell/bulk preparation gap without erasing the signal that matters for prediction, and whether grounding the task loss in real biology works better than a generic classifier would.

Uncertainties remain: GAN training is known to be unstable in general, and whether that holds at this sample size and dimensionality (∼840 pairs, 5000 dimensions) is untested. PBMCs completely lack granulocytes, requiring $G_{S\to B}$ to reconstruct granulocyte-associated gene expression from how other genes co-vary alone. This approach also relies on same-draw pairing for ground-truth composition, unavailable in our broader study's unmatched cohorts.

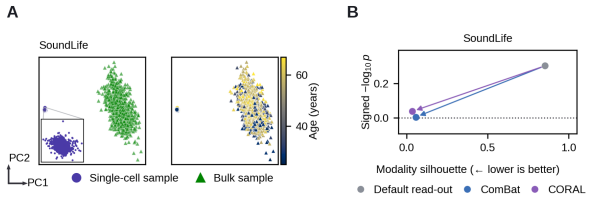


Figure 1. (a) Joint PCA of CytoSet's embedding for single-cell and bulk samples (AIFI Sound Life), colored by modality (single-cell samples enlarged in an inset) and by age. PC1 separates the assays, not age (silhouette 0.85, PC1 η^2 modality 0.98, age < 0.01). (b) Modality silhouette of the embeddings in (a) against young-vs-old transfer (signed $-\log_{10} p$) for the default read-out (gray; its LASSO z-scores each assay on its own) and after ComBat or CORAL: both corrections collapse the silhouette from 0.85 to 0.06 and 0.04 without recovering transfer, which stays at chance.

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